

SWE363 | Web Engineering & Development

Project | Phase Three

Your Gym AI | Team 08 | section 01



Team Members

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1. Motivation

Your Gym AI is a fitness platform that utilizes AI to provide personalized training and nutrition programs. It fills the gap between generic, one-size-fits-all programs and individual coaching from a human coach, which is typically pricey, limited in terms of time, and hard to find. The AI-generated programs adapt progressive planning using GPT-style generation, while still allowing each member to achieve their goals. The problem matters to be solved because most beginners quit a fitness program in a few weeks because they do not have a clear routine or adherence to an exercise program.

2. Target Users

- Guest: Unauthenticated visitor who can browse the website, preview features, and create an account.
- Member (User): Registered user who completes onboarding, receives AI-generated workouts/meals, tracks progress, and follows a weekly schedule.
- Coach: Certified professional who manages assigned clients, edits workouts/meals, and leaves comments to users.
- Admin: System administrator who manages users and coaches, assigns relationships, monitors platform metrics, and enforces policy.

3. Requirements Modeling

3.1 Functional Requirements

3.1.1. Guests:

FR-1) Browse Website:

Normal flow:

1. Guest visits the homepage.
2. System displays feature highlights and call-to-action buttons.
3. Guest navigates sections without logging in.

Validation points:

- No auth required.

System responses:

- Display marketing content; no data stored.

FR-2) Create Account:

Normal flow:

1. Guest clicks "Sign Up".
2. System shows registration form (email, password, name).
3. Guest submits form.
4. System validates email uniqueness and password policy.
5. On success, user recorded and session/token created; welcome message shown.

Validation points:

- Unique email; password length ≥ 6 ; trimmed/lower-cased email.

System responses:

- Persist user in DB; issue JWT; return success JSON and redirect to onboarding.

3.1.2. Users:

FR-3) Login / Logout:

Normal flow:

1. User opens login page and submits credentials.
2. System validates credentials and rate limits excessive attempts.
3. If valid, token is generated and session begins.
4. User logs out; token/session invalidated.

Validation points:

- correct email/password; account active.

System responses:

- Tokens returned; 429 if too many attempts; 401 if invalid credentials.

FR-4) Complete Onboarding:

Normal flow:

1. User provides profile and preferences (age, height, goals, equipment).
2. System saves data and generates initial workout and nutrition plans.
3. Plans stored and user redirected to dashboard.

Validation points:

- All required fields must be filled and valid.

System responses:

- User record updated; plan data stored in DB.

FR-5) View Weekly Schedule / Today's Workout:

Normal flow:

1. User opens the Schedule tab.
2. System retrieves weekly plan and today's workout.
3. User selects workout to view details.

Validation points:

- User must be authenticated.

System responses:

- Display list of workouts and details from DB.

FR-6) Track Workout Progress:

Normal flow:

1. User marks exercises as complete during workout.
2. System updates progress metrics in real-time.
3. Progress bar and completion percentage are shown.

Validation points:

- User must be authorized for that workout.

System responses:

- Workout progress updated in DB; confirmation displayed.

3.1.3. Coaches:

FR-7) Manage Client Workouts:

Normal flow:

1. Coach views list of assigned clients.
2. Coach selects a client and opens their workout plan.
3. Coach edits or adds workouts and saves updates.

Validation points:

- Coach must be assigned to that client.

System responses:

- Workout plan updated; confirmation displayed.

FR-8) Add Comments and Regenerate Plan:

Normal flow:

1. Coach adds feedback or comments to a client's exercise or meal.
2. Coach optionally regenerates client plan using AI assistance.

Validation points:

- Rate-limit AI regeneration; ensure assigned relationship.

System responses:

- New plan version stored; notification sent to client.

3.1.4. Admins:

FR-9) Manage Users and Coaches:

Normal flow:

1. Admin logs in and opens dashboard.
2. System displays all users and coaches.
3. Admin views, edits, or deletes user accounts.

Validation points:

- Admin role required; cannot delete self.

System responses:

- Action recorded; changes reflected immediately.

FR-10) Assign Coaches to Clients:

Normal flow:

1. Admin selects client and available coach.
2. Admin confirms assignment.
3. System links the client with the coach.

Validation points:

- Both IDs valid; prevent duplicate mapping.

System responses:

- Mapping stored; success confirmation shown.

3.2 Non-Functional Requirements

3.2.1 Performance:

- Support 500 concurrent members with horizontal scaling of Node workers and DB indexes.

3.2.2 Security:

- role-based authorization (member/coach/admin).
- Strong password policy by Bcrypt hashing.
- Input validation and sanitization; rate-limiting on auth and AI routes; audit logs for admin actions.

3.2.3 Usability & Accessibility:

- Responsive UI across mobile/desktop.

3.2.4 Compatibility:

- role Modern browsers such as Chrome, Edge, Firefox, and Safari.

3.2.5 External Constraints:

- OpenAI API for plan/chat; MongoDB for persistence.
- Node.js/Express backend; React + Vite frontend.

4. Wireframes

Figma Link: <https://www.figma.com/design/mL3Z7r32zACxIClGFqBRL2/SWE-363?node-id=0-1&t=p6tZiJNGrCI0tNW6-1>